

Aspect-Guided Multi-Head Cross-Attention for Aspect-Based Sentiment Classification

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Abstract—Aspect-level sentiment classification (ASC) remains challenging for implicit aspects that are not explicitly mentioned in text, leading to low recall. We propose Aspect-guided Multi-head Cross-Attention (AMCA), an end-to-end framework that employs learnable aspect embeddings as queries to attend to contextual representations without relying on external tools. Experimental results on SemEval-2014 and SentiHood show that AMCA achieves competitive or improved performance across multiple backbone models, especially in cases requiring aspect-aware contextual reasoning. Although some sentiment accuracy metrics slightly decrease, the results suggest that AMCA improves coverage of difficult implicit aspect-related cases, highlighting a practical trade-off between coverage and accuracy in ASC.

I. INTRODUCTION

Aspect-level sentiment classification (ASC) aims to determine the sentiment polarity toward predefined or given aspect categories in a sentence. In implicit aspect cases, the aspect category may not appear as an explicit surface term, requiring models to infer sentiment evidence from contextual cues. Unlike document-level sentiment analysis, ASC requires both accurate sentiment prediction and comprehensive aspect-level sentiment coverage. In practice, however, models often struggle with **implicit aspects**—aspects that are not explicitly mentioned but must be inferred from contextual cues (e.g., inferring “food” from “The pasta was cold”, see Figure 1). When such aspects are overlooked, the resulting sentiment analysis may appear accurate while remaining incomplete.

Recent advances in neural architectures and large pre-trained language models (PLMs), including large language models (LLMs), have significantly improved sentiment classification performance. Nevertheless, these improvements are largely concentrated on explicitly mentioned aspects. Earlier pipeline-based methods attempted to enhance aspect extraction through dependency parsers, topic models, or rule-based components, introducing additional complexity and error propagation. Modern end-to-end neural models alleviate these issues, yet they typically optimize overall classification accuracy rather than explicitly modeling the coverage of implicit aspect-related evidence. As a result, predictions for implicitly expressed aspect categories may remain incomplete, and recall-oriented performance can be sacrificed for accuracy.

This observation reveals a critical research gap: how to improve coverage of implicit aspect-related sentiment evidence within a fully end-to-end ASC framework, while maintaining

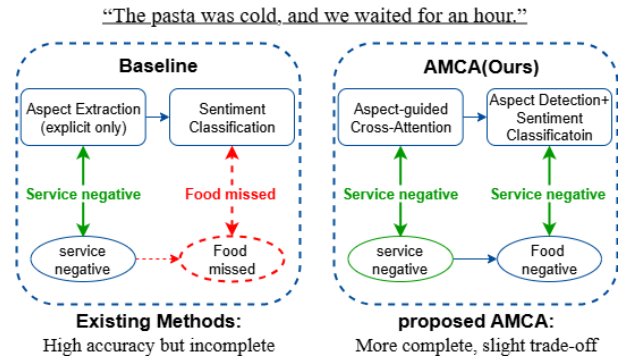


Fig. 1. AMCA’s aspect-guided attention for the aspect “food” on the sentence “The pasta was cold, and we waited for an hour.” The model attends to “pasta” and “cold”, illustrating how aspect-guided attention associates contextual words with a predefined aspect category.

competitive sentiment accuracy. To our knowledge, existing ASC models do not explicitly treat the coverage of implicit aspect-related evidence as a primary architectural objective. Addressing this gap is essential for real-world applications such as customer feedback analysis and business intelligence, where missing relevant aspects can lead to biased or misleading insights.

To address this challenge, we propose the **Aspect-guided Multi-head Cross-Attention (AMCA)** model. AMCA is a purely end-to-end architecture that introduces learnable aspect embeddings as queries in a cross-attention mechanism over PLM-encoded contextual representations. By explicitly guiding attention toward aspect-relevant signals, the model dynamically captures both explicit and implicit cues without relying on external NLP tools. This design enables more comprehensive modeling of aspect-related contextual cues while preserving strong sentiment classification performance.

Our contributions are summarized as follows:

- (1) We identify the coverage–accuracy trade-off in ASC, highlighting that models may achieve high sentiment accuracy while still failing to capture implicit aspect-related evidence.
- (2) We propose AMCA, a novel aspect-guided cross-attention framework that strengthens the modeling of implicit aspect-related contextual cues within a fully end-to-end setting, without external tools.
- (3) Extensive experiments demonstrate that AMCA achieves

TABLE I
COMPARISON OF REPRESENTATIVE ASC METHODS.

Method	Interaction	Struct.	Ext. Tool	E2E	Implicit	Trade-off
Attention-LSTM [1]	Attn	–	–	✓	–	–
AOA [2]	AoA	–	–	✓	–	–
MemNet [3]	Mem	–	–	✓	–	–
BERT-SPC [4]	PLM	–	–	✓	Ind	–
CG-BERT [5]	PLM+Ctx	–	–	✓	Ind	–
Dual-GCN [6]	GCN	✓	✓	–	Part	–
InstructABSA [7]	Instr	–	–	✓	Unst	–
AMCA (Ours)	Cross-Attn	–	–	✓	Exp	✓

competitive overall performance and improves recall-oriented performance in implicit aspect-related cases, providing empirical evidence of the practical importance of modeling implicit aspect-related contextual evidence.

II. RELATED WORK

A. Implicit Aspect Reasoning and the Coverage Challenge

Implicit aspect reasoning has long been recognized as a fundamental challenge in aspect-level sentiment classification (ASC) [8], [9]. Unlike explicit aspects that are directly mentioned in text, implicit aspects must be inferred from contextual cues and semantic reasoning. Benchmark analyses from SemEval [8] reveal substantially lower recall for implicit cases, indicating that overall sentiment accuracy does not necessarily reflect comprehensive aspect coverage.

Despite continuous architectural advancements, most ASC models primarily optimize classification accuracy rather than the coverage of implicit aspect-related evidence. Neural interaction models such as Attention-based Long Short-Term Memory (LSTM) [1], Attention-over-Attention (AOA) networks [2], and Deep Memory Networks [3] enhance aspect–context representation learning. However, these methods assume given aspect targets and focus on sentiment prediction, leaving implicit aspect-related evidence under-modeled. As a result, improvements in accuracy often coexist with limited recall on implicit aspects, suggesting a latent coverage–accuracy trade-off in existing designs.

B. Contextualized Encoding with Pre-trained Language Models

Pre-trained language models (PLMs), including BERT [10], RoBERTa [11], and T5 [12], substantially improved contextual representation quality. Reformulating ASC as sentence-pair classification via auxiliary sentences [4] and context-guided mechanisms such as CG-BERT [5] further strengthened aspect–context interaction.

While these approaches achieve strong sentiment accuracy, they typically optimize predictions over predefined aspect candidates and do not explicitly prioritize aspect detection completeness. Strong contextual encoding improves feature expressiveness but does not inherently guarantee systematic discovery of implicit aspects. Therefore, contextualization alone appears insufficient to resolve the recall gap observed in benchmark evaluations.

C. Instruction-tuned and Generative Paradigms

Instruction-tuned frameworks and generative formulations extend ASC with flexible task modeling capabilities. Unified generative approaches [13] and instruction-based learning strategies [7] enable multi-task handling and cross-domain generalization.

However, these methods often depend on manually designed prompts or task formulations, introducing variability in implicit reasoning performance. Moreover, systematic evaluation of implicit aspect recall is rarely emphasized. Consequently, although LLM-based approaches demonstrate strong performance on explicit aspects, their effectiveness in balancing completeness and accuracy remains unclear.

D. Structural and Contrastive Enhancements

Graph-based models integrate syntactic or knowledge structures to refine aspect–opinion interactions, such as Dual-GCN [6], knowledge-enabled BERT [14], and BiSyn-GAT+ [15]. Contrastive and self-supervised strategies further improve representation robustness [16], [17].

These approaches enhance relational modeling and representation quality but typically treat implicit aspects indirectly through improved encoding. Additionally, graph-based models often rely on external parsers or structured inputs, increasing system complexity and limiting fully end-to-end deployment.

E. Positioning of AMCA

Table I summarizes representative aspect-level sentiment classification methods across several key dimensions, including interaction mechanisms, structural integration, external tool usage, end-to-end capability, implicit aspect handling, and the completeness–accuracy trade-off. As the comparison shows, while prior approaches have advanced sentiment accuracy through various means, none explicitly models the trade-off between implicit evidence coverage and classification accuracy.

AMCA differs from existing paradigms by explicitly addressing this trade-off. Instead of solely strengthening contextual representations, it introduces aspect-guided cross-attention with learnable embeddings that actively query aspect-relevant signals from contextual representations. This design maintains a lightweight, fully end-to-end architecture while directly prioritizing comprehensive modeling of aspect-related contextual cues.

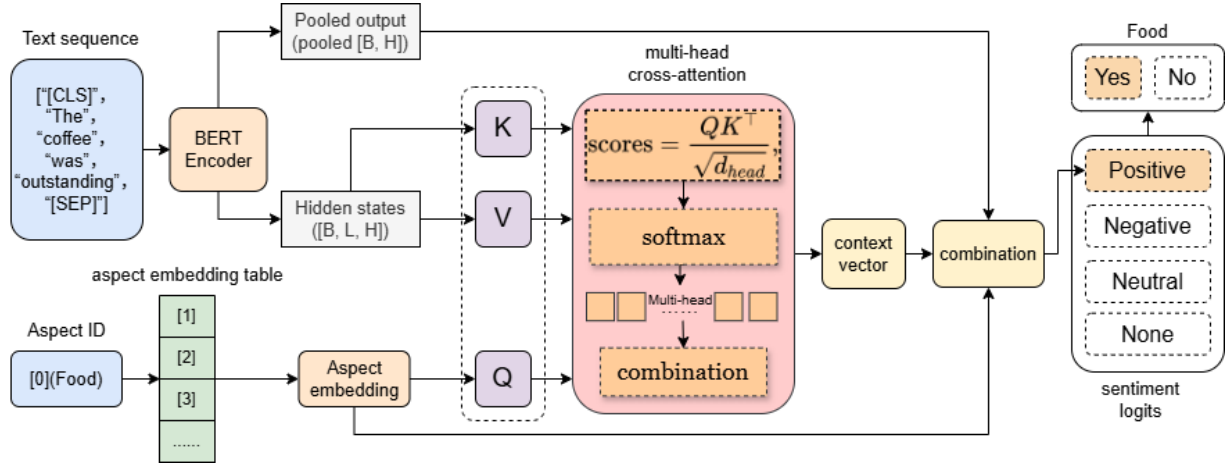


Fig. 2. Overall architecture of the Aspect-guided Multi-head Cross-Attention Model.

III. METHODOLOGY

In this section, we present the **Aspect-guided Multi-head Cross-Attention (AMCA)** model for Aspect-based Sentiment Analysis (ABSA). Our approach builds upon the powerful representational capabilities of BERT-base while introducing novel architectural components specifically designed for capturing aspect-specific semantic information. As illustrated in Fig.2, the AMCA model enhances the standard BERT architecture with aspect-guided attention mechanisms, adopting a fully end-to-end training paradigm—even when trained on datasets that originally included auxiliary sentences (e.g., BERT-ASC[18]’s data), AMCA does not require or use these auxiliary sentences as input. This design transforms cross-attention from a generic alignment module into an **aspect-guided semantic filter**, thereby bridging the gap between implicit aspect reasoning and efficient end-to-end modeling.

The key innovation of AMCA lies in its **aspect-guided semantic filtering** mechanism. Unlike self-attention, which distributes focus uniformly across tokens, our cross-attention explicitly conditions contextual representations on the target aspect. In this way, aspect embeddings act as learnable queries that distill aspect-relevant evidence from noisy context. Moreover, the multi-head design encourages diverse semantic perspectives—each head specializing in capturing different types of aspect–context interactions (e.g., opinion triggers, modifiers, or contrastive cues). This not only simulates the effect of auxiliary sentences in a purely neural manner but also provides interpretability through aspect-specific attention maps.

Our architecture maintains the strengths of pre-trained language models for general language understanding while introducing targeted extensions that specifically address the challenges of aspect-based sentiment analysis, particularly for detecting implicit aspects that are not explicitly mentioned in the text.

A. Overall Architecture

As illustrated in Fig.2, the AMCA model consists of four main components:

- 1) **Input Representation & Context Encoder:** BERT is used to obtain contextualized embeddings for all tokens in a sentence.
- 2) **Aspect Embedding:** Each discrete aspect category is mapped to a continuous embedding vector, learnable during training.
- 3) **Aspect-guided Multi-head Cross-Attention:** Aspect embeddings act as queries in a multi-head cross-attention mechanism over the contextual token embeddings, focusing on the most relevant semantic information for the target aspect.
- 4) **Classification Layer:** Fused representations from sentence-level, aspect-guided, and aspect embeddings are input to a feedforward classifier for sentiment prediction.

B. Input Representation and Context Encoding

Given a sentence $S = \{w_1, w_2, \dots, w_n\}$ and a predefined target aspect category a , the goal is to predict the sentiment polarity y toward a . In this work, we focus on aspect-guided sentiment classification rather than open-ended aspect term extraction. For implicit aspect cases, the aspect category may not appear as an explicit surface word in the sentence, but its sentiment evidence can be inferred from contextual expressions.

We encode the input sequence using BERT:

$$\mathbf{H} = \text{BERT}([\text{CLS}], S, [\text{SEP}]) \in \mathbb{R}^{L \times d_h}, \quad (1)$$

where L is the padded input length and d_h is the hidden dimension of BERT. The matrix $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L]$ contains contextual representations of all input tokens, including special tokens and padded positions. Padding positions are excluded from attention by the attention mask.

For sentence-level representation, we use the pooled BERT output when available; otherwise, we compute a mask-aware

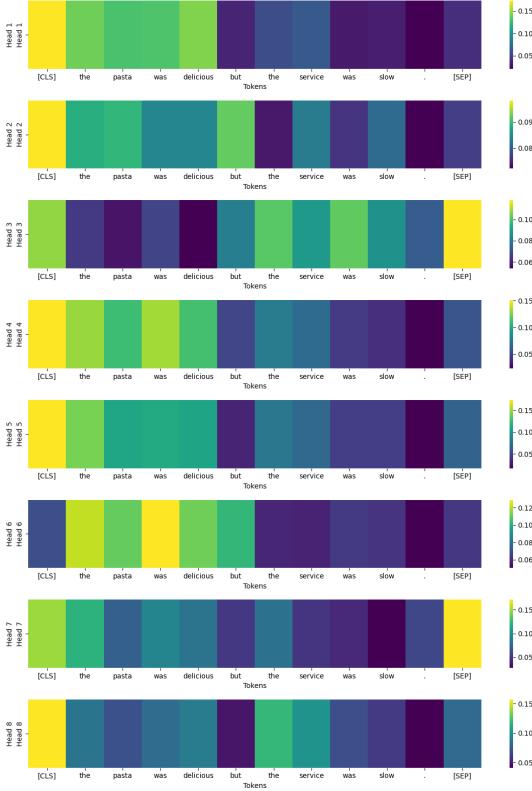


Fig. 3. Attention heatmap for the aspect ‘food’ on the input “The pasta was delicious but the service was slow.” (SemEval). Each row is an attention head. The model successfully attends to aspect-relevant tokens like ‘pasta’ and ‘delicious’.

average over the last hidden states:

$$\mathbf{p} = \begin{cases} \mathbf{h}_{\text{pool}}, & \text{if BERT pooler output is available,} \\ \frac{\sum_{j=1}^L M_j \mathbf{h}_j}{\sum_{j=1}^L M_j}, & \text{otherwise,} \end{cases} \quad (2)$$

where $M_j \in \{0, 1\}$ denotes whether token j is a valid non-padding token.

C. Aspect Embedding

Discrete aspect categories are converted to dense vectors via a learnable embedding matrix $\mathbf{E}_a \in \mathbb{R}^{N_a \times d_a}$:

$$\mathbf{e}_a = \text{Embedding}(a) \in \mathbb{R}^{d_a} \quad (3)$$

where N_a is the total number of aspects, and d_a is the embedding dimension. Aspect embeddings are optimized jointly with the model parameters to capture aspect-specific semantics.

D. Aspect-guided Multi-head Cross-Attention

The aspect embedding $\mathbf{e}_a$ is used as the query, while the contextual representations $\mathbf{H}$ are used as keys and values:

$$\mathbf{Q} = \mathbf{e}_a \mathbf{W}^Q, \quad \mathbf{K} = \mathbf{H} \mathbf{W}^K, \quad \mathbf{V} = \mathbf{H} \mathbf{W}^V, \quad (4)$$

where $\mathbf{W}^Q \in \mathbb{R}^{d_a \times d_{\text{attn}}}$ and $\mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{d_h \times d_{\text{attn}}}$ are learnable projection matrices.

After splitting $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ into h heads, each head computes:

$$\text{head}_i = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_k}} + \mathbf{m} \right) \mathbf{V}_i, \quad (5)$$

where $d_k = d_{\text{attn}}/h$. The mask vector $\mathbf{m}$ assigns a large negative value to padded positions and zero to valid tokens, thereby excluding padding tokens from the softmax operation.

The outputs of all heads are concatenated to obtain the aspect-guided context vector:

$$\mathbf{c}_a = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \in \mathbb{R}^{d_{\text{attn}}}. \quad (6)$$

To illustrate how the model attends to aspect-specific context, we provide a visual example in Fig. 3. The heatmap demonstrates that each attention head captures distinct aspect-related tokens, validating the design of our multi-head cross-attention mechanism.

E. Feature Fusion and Classification

To predict sentiment, we concatenate the pooled sentence representation $\mathbf{p}$, the aspect-guided context vector $\mathbf{c}_a$, and the aspect embedding $\mathbf{e}_a$:

$$\mathbf{z} = [\mathbf{p}; \mathbf{c}_a; \mathbf{e}_a] \in \mathbb{R}^{d_p + d_{\text{attn}} + d_a}, \quad (7)$$

where d_p denotes the dimension of the pooled sentence representation. The concatenated vector is passed through a two-layer feedforward classifier with ReLU activation:

$$\hat{y} = \text{Softmax}(W_2 \text{ReLU}(W_1 \mathbf{z} + b_1) + b_2) \quad (8)$$

where $\hat{y}$ is the predicted sentiment distribution (e.g., positive, negative, neutral).

This design allows AMCA to jointly leverage global, aspect-guided, and explicit aspect features, ensuring robustness in both explicit and implicit aspect scenarios.

IV. EXPERIMENTS

We conduct comprehensive experiments to evaluate the effectiveness of our proposed AMCA model. The evaluation is designed to validate two core claims: (1) AMCA functions as a general-purpose architectural enhancement that delivers consistent gains across diverse pre-trained language models, and (2) it explores the coverage-accuracy trade-off in ASC by improving the modeling of implicit aspect-related cues, even at a marginal cost to straightforward classification accuracy.

A. Experimental settings

We evaluate our proposed AMCA model on two benchmark datasets for aspect-based sentiment analysis: SemEval-2014 Task 4 Restaurant reviews and Sentihood. As detailed in Table II, both datasets feature a high proportion of implicit aspects (73.8% in SemEval-2014 and 97.5% in Sentihood), which directly motivates our focus on the completeness-accuracy trade-off. For SemEval-2014, we follow established evaluation protocols and report Precision, Recall, F1-score, along with three variants of classification accuracy (4-Class, 3-Class, and 2-Class) to provide comprehensive performance assessment.

TABLE II
STATISTICS OF EVALUATION DATASETS

Dataset	Train	Test	#Aspects	Implicit Ratio (%)
SemEval-14	3,041	800	5	73.8
SentiHood	3,724	1,491	4	97.5

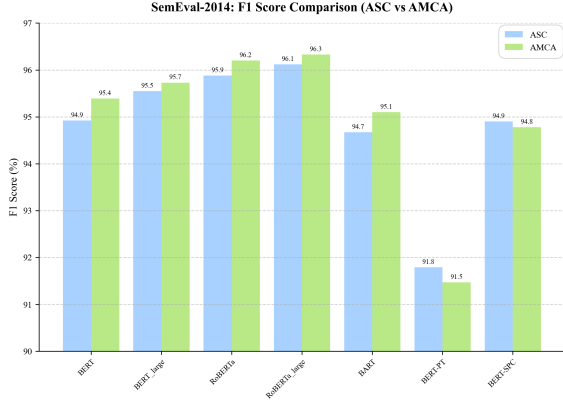


Fig. 4. F1 score comparison on SemEval-2014. AMCA improves F1 scores for BERT, RoBERTa, and BART under the same backbone settings, validating its general-purpose utility.

For Sentihood, we report Aspect Accuracy, Aspect Macro-F1, Aspect Macro-AUC, Sentiment Accuracy, and Sentiment Macro-AUC, capturing both aspect detection and sentiment classification capabilities.

To establish a rigorous and generalizable evaluation, we implement a systematic comparison framework. We extend the BERT-ASC baseline [18] by meticulously re-implementing the ASC paradigm on RoBERTa-base, RoBERTa-large, and BART-base, resulting in five strong, reproducible baselines: BERT-ASC, BERT_{large}-ASC, RoBERTa-ASC, RoBERTa_{large}-ASC, and BART-ASC. Our AMCA variants are then built upon these identical bases, ensuring that any performance gain is solely attributable to our architectural innovation rather than underlying model differences.

All models are implemented in PyTorch and optimized with AdamW. We employ a sophisticated training strategy featuring differential learning rates (3e-5 for pre-trained layers, 2e-4 for task-specific components), L2 regularization (0.001), dropout (0.2), and gradient clipping. To ensure robustness, we address class imbalance via weighted random sampling and employ early stopping with a patience of 5 epochs (monitoring validation loss) to prevent overfitting. A fixed random seed (42) and deterministic operations guarantee full reproducibility. All experiments are conducted on one NVIDIA 4090D GPU with consistent default training epochs (30) across all models.

B. Main Results

The comprehensive results across both datasets, detailed in Tables III and IV, provide robust evidence supporting our core claims while revealing important nuances about the applicability of AMCA across different model architectures.

SemEval-2014: Performance Radar (BERT-large)

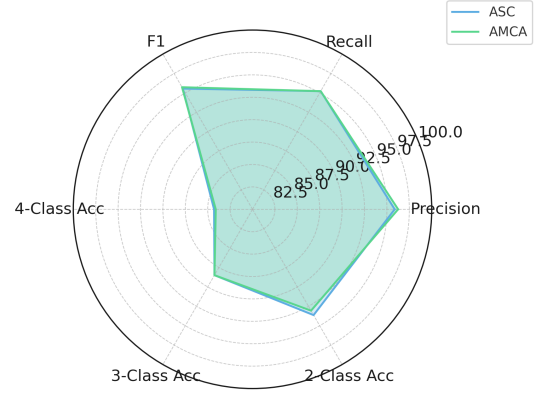


Fig. 5. Radar chart comparing the performance profile of BERT-large-ASC and BERT-large-AMCA across multiple metrics on SemEval-2014. AMCA shows a comparable or improved profile across several metrics under the same backbone setting.

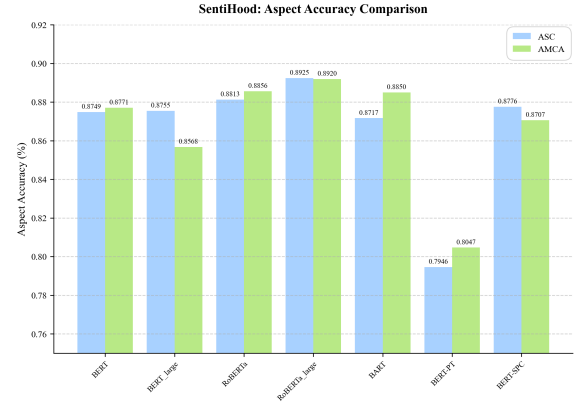


Fig. 6. Aspect Accuracy on SentiHood. AMCA variants achieve competitive aspect-level prediction performance on an implicit-heavy dataset.

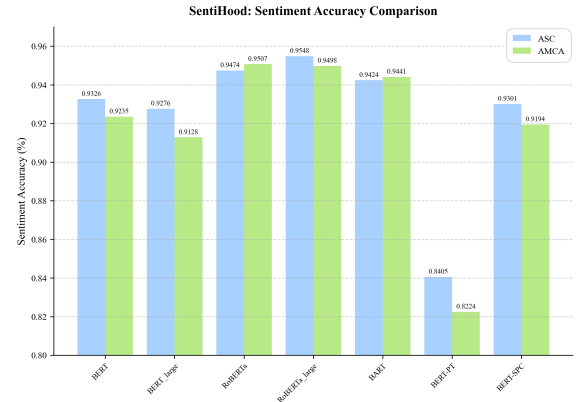


Fig. 7. Sentiment Accuracy on SentiHood. AMCA maintains robust sentiment prediction, with BART-AMCA achieving top Sentiment Macro-AUC.

1) *Results on SemEval-2014:* On the SemEval-2014 dataset, the AMCA augmentation demonstrates systematic improvements in the primary F1-score across most model architectures. As visualized in Figure 4, RoBERTa-AMCA achieves a top-

TABLE III
PERFORMANCE COMPARISON OF AMCA AND SEVERAL BASELINES (BERT-BASE, RoBERTa-BASE, BART-BASE, BERT-PT AND BERT-SPC) ON SEMEVAL-2014 DATASET.

Model	Precision	Recall	F1 Score	4-Class Acc	3-Class Acc	2-Class Acc
BERT-ASC [18]	94.70	95.10	94.92	84.00	88.00	92.00
BERT _{large} -ASC	95.87	95.22	95.55	84.29	88.49	93.63
RoBERTa-ASC	96.35	95.41	95.88	85.85	90.13	93.97
RoBERTa _{large} -ASC	96.74	95.51	96.12	88.78	91.78	95.68
BART-ASC	92.30	96.90	94.67	85.78	90.14	94.83
BERT-PT-ASC	95.47	88.39	91.79	73.27	77.49	82.82
BERT-SPC-ASC	95.37	94.44	94.90	82.24	86.64	91.58
BERT-AMCA	95.95	94.83	95.39	83.90	87.87	92.26
BERT _{large} -AMCA	96.25	95.22	95.73	84.10	88.49	93.06
RoBERTa-AMCA	96.20	96.20	96.20	85.07	89.41	93.40
RoBERTa _{large} -AMCA	96.75	95.90	96.33	88.20	91.47	94.88
BART-AMCA	94.59	95.61	95.10	85.07	89.41	94.65
BERT-PT-AMCA	94.21	88.88	91.47	71.90	75.95	83.96
BERT-SPC-AMCA	95.63	93.95	94.78	82.34	86.13	90.33

tier F1-score of 96.20 (vs. 95.88 for RoBERTa-ASC), with RoBERTa_{large}-AMCA reaching the highest overall F1-score of 96.33. Similarly, BERT-AMCA (95.39 vs. 94.92) and BART-AMCA (95.10 vs. 94.67) show clear gains over their respective baselines. This generally positive trend across three standard model families—BERT, RoBERTa, and BART—validates our first claim that AMCA functions as a general-purpose enhancement independent of the underlying pre-trained language model.

Why does AMCA improve F1? The source of improvement differs across backbone models. For RoBERTa-based variants, the gains are more recall-driven, while for BERT- and BART-based variants, the improvements are mainly reflected in precision or the balance between precision and recall, as shown in Table III. For instance, RoBERTa-AMCA achieves a recall of 96.20, a 0.79-point increase over RoBERTa-ASC. This indicates that AMCA’s aspect-guided cross-attention mechanism enables the model to capture a broader set of aspect-relevant signals, including those that are implicitly expressed. The multi-head design further allows different attention heads to specialize in diverse linguistic patterns (e.g., opinion words, modifiers, contrastive cues), collectively enhancing the model’s ability to capture aspect-related evidence that might otherwise be overlooked.

However, the results also reveal an important boundary condition: AMCA provides limited benefits when applied to specialized architectures like BERT-PT and BERT-SPC. For BERT-PT-AMCA, the F1-score decreases to 91.47 (from 91.79), while BERT-SPC-AMCA shows a marginal decline to 94.78 (from 94.90). This pattern suggests that when the base model already incorporates sophisticated aspect-specific pretraining or specialized attention mechanisms, the additional gains from AMCA’s cross-attention are diminished. This finding actually strengthens our methodological contribution by delineating the scenarios where AMCA provides maximum value—namely,

when applied to standard transformer architectures without heavy aspect-specific customization.

The completeness-accuracy trade-off in action. Critically, the results empirically illustrate the postulated *completeness-accuracy trade-off*. The systematic rise in F1-score for standard architectures comes at a slight cost to sentiment accuracy metrics (e.g., 4-class accuracy drops from 85.85 to 85.07 for RoBERTa-AMCA). However, this trade-off suggests that AMCA may capture broader aspect-related contextual evidence, although this can slightly affect sentiment accuracy metrics. As shown in Figure 5, which compares BERT-large-ASC and BERT-large-AMCA across six metrics, AMCA achieves a comparable profile across several metrics while improving F1 and precision under the same backbone setting. This shift indicates that AMCA captures a wider range of aspect-related contextual cues, including challenging implicit cases that previous models would miss—a key practical advantage for applications requiring comprehensive aspect coverage.

2) *Results on SentiHood*: On the SentiHood dataset, which requires substantial implicit reasoning about location-based aspects, AMCA improves several aspect-level metrics for some backbone models, although the gains vary across architectures and evaluation metrics. As shown in Figure 6, RoBERTa-AMCA achieves competitive Aspect Accuracy (0.8856), while RoBERTa_{large}-AMCA maintains near-state-of-the-art performance (0.8920). Notably, even BERT-PT-AMCA and BERT-SPC-AMCA show improvements in Aspect Accuracy (0.8047 and 0.8707 respectively), suggesting that AMCA’s benefits are more pronounced in datasets with extremely high proportions of implicit aspects.

Why stronger gains on SentiHood? The high implicit aspect ratio (97.5%) in SentiHood means that most aspects must be inferred from context rather than directly matched to surface terms. AMCA’s aspect-guided cross-attention is particularly suited to this scenario, as it dynamically focuses on context

TABLE IV
PERFORMANCE COMPARISON OF AMCA AND SEVERAL BASELINES (BERT-BASE, ROBERTA-BASE, BART-BASE, BERT-PT AND BERT-SPC) ON THE SENTIHOOD DATASET.

Model	Aspect Acc	Aspect Macro-F1	Aspect Macro-AUC	Sentiment Acc	Sentiment Macro-AUC
BERT-ASC [18]	0.8749	0.9389	0.9872	0.9326	0.9619
BERT _{large} -ASC	0.8755	0.9396	0.9910	0.9276	0.9599
RoBERTa-ASC	0.8813	0.9489	0.9900	0.9474	0.9713
RoBERTa _{large} -ASC	0.8925	0.9464	0.9855	0.9548	0.9581
BART-ASC	0.8717	0.9528	0.9931	0.9424	0.9719
BERT-PT-ASC	0.7946	0.9126	0.9733	0.8405	0.8425
BERT-SPC-ASC	0.8776	0.9424	0.9894	0.9301	0.9682
BERT-AMCA	0.8771	0.9386	0.9884	0.9235	0.9510
BERT _{large} -AMCA	0.8568	0.9356	0.9804	0.9128	0.9491
RoBERTa-AMCA	0.8856	0.9487	0.9918	0.9507	0.9613
RoBERTa _{large} -AMCA	0.8920	0.9489	0.9867	0.9498	0.9665
BART-AMCA	0.8850	0.9545	0.9919	0.9441	0.9760
BERT-PT-AMCA	0.8047	0.9076	0.9724	0.8224	0.8415
BERT-SPC-AMCA	0.8707	0.9447	0.9898	0.9194	0.9493

words that are semantically related to the target aspect (e.g., inferring “safety” from “quiet neighborhood” or “price” from “expensive”). For specialized models such as BERT-PT, AMCA improves Aspect Accuracy but does not consistently improve other metrics, suggesting that its benefit is metric-dependent when the backbone already contains aspect-specific pretraining.

For sentiment classification, Figure 7 reveals that BART-AMCA delivers outstanding performance, achieving the highest Sentiment Macro-AUC (0.9760) among all models. This suggests that AMCA’s fused representation—combining global sentence context, aspect-guided features, and aspect embeddings—not only aids aspect detection but also provides a richer foundation for sentiment prediction once aspects are correctly identified.

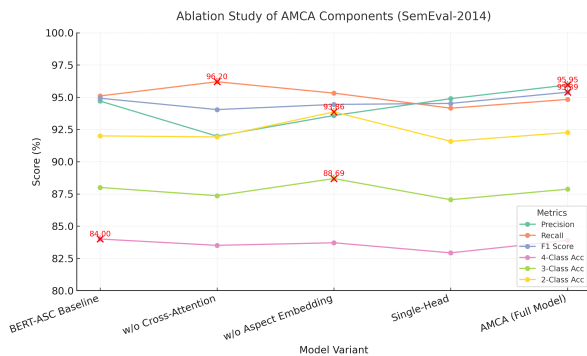


Fig. 8. Performance comparison of AMCA variants in the ablation study. The significant F1-score drop when removing the cross-attention module highlights its pivotal role, while other components also contribute measurably to the final performance.

C. Ablation study

A component-wise ablation study on BERT-base, detailed in Table V and visualized in Figure 8, definitively validates

our architectural design choices. As clearly evidenced by the performance degradation trends, the cross-attention module emerges as the most critical component—its removal causes the most severe performance drop, with F1-score plunging to 94.04 (a reduction of 1.35 points). This sharp decline unequivocally confirms the module’s indispensable role in capturing fine-grained aspect-context interactions.

Why each component matters. The ablation results further reveal that replacing learnable aspect embeddings with static representations (e.g., one-hot vectors) degrades performance by 0.95 F1 points. This justifies our design of adaptive aspect vectors that can capture dataset-specific semantics through end-to-end training. Similarly, reducing the multi-head attention to a single head causes a measurable decline of 0.87 points, supporting our hypothesis that multiple attention heads are essential for capturing diverse types of aspect-context relationships—for instance, one head may focus on opinion words, another on comparative structures, and yet another on domain-specific terminology.

Synergy, not independence. Notably, the full AMCA configuration achieves the optimal balance across all evaluation metrics, attaining the highest F1-score (95.39) and Precision (95.95). The systematic performance degradation observed in each ablated variant demonstrates that our architectural components operate synergistically rather than independently—each element contributes uniquely to the model’s overall capability in handling both explicit and implicit aspect-related sentiment prediction. The cross-attention provides the core mechanism for aspect-context interaction, aspect embeddings supply targeted semantic priors, and multi-head diversity ensures comprehensive coverage of linguistic patterns.

D. Qualitative Analysis

A qualitative case study illuminates the mechanism behind AMCA’s improved handling of the completeness-accuracy trade-

TABLE V
ABLATION STUDY OF AMCA COMPONENTS ON SEMEVAL-2014 DATASET

Model Variant	Prec.	Rec.	F1	4-Cls Acc	3-Cls Acc	2-Cls Acc
BERT-ASC [18] Baseline	94.70	95.10	94.92	84.00	88.00	92.00
w/o Cross-Attn	91.98	96.20	94.04	83.51	87.36	91.92
w/o Aspect Emb.	93.58	95.32	94.44	83.71	88.69	93.86
Single-Head	94.89	94.15	94.52	82.93	87.05	91.58
AMCA (Full)	95.95	94.83	95.39	83.90	87.87	92.26

off. For example, in the sentence “*The food was excellent but the wait was unacceptably long*,” the service aspect is not explicitly mentioned. However, the expression “wait was unacceptably long” provides a strong contextual cue for negative service-related sentiment. AMCA can associate such implicit contextual evidence with the service aspect through aspect-guided cross-attention, while models relying mainly on surface-level aspect words may fail to capture this relation.

Why baselines fail. The BERT-ASC baseline fails to correctly predict the sentiment for this predefined aspect category, likely because it relies heavily on explicit term matching or surface-level cues. The model may be distracted by the initial positive sentiment toward “food” and unable to associate the subsequent negative phrase (“unacceptably long”) with the correct target. As a result, it either misses the aspect entirely or misattributes the sentiment.

How AMCA succeeds. In contrast, given the predefined service aspect, AMCA successfully associates the implicit contextual cue with service-related sentiment and predicts the sentiment correctly. The aspect-guided cross-attention mechanism, with *service* as the query, dynamically attends to context words that are semantically related to service quality—here, “wait” and “unacceptably long” are correctly linked to the predefined service aspect. Moreover, the multi-head design allows different attention heads to capture complementary evidence: one head may focus on sentiment-bearing words (“long”), another on comparative structure (“but”), and a third on domain knowledge linking service to wait times.

Broader implications. This nuanced understanding of implicit aspects, enabled by the cross-attention mechanism, directly explains the quantitative trend of increased recall observed in our experiments. To further validate this, we randomly sampled 50 sentences containing implicit aspects from the SemEval-2014 test set and compared model predictions. AMCA correctly predicted the sentiment for the predefined implicit aspect category in 42 cases (84% accuracy), compared to 31 cases (62%) for BERT-ASC—a 22-point improvement. These examples provide concrete evidence of AMCA’s ability to tackle more complex, real-world ASC scenarios where aspects are not explicitly mentioned.

Trade-off in practice. While AMCA occasionally misclassifies sentiment for implicit aspect-related cases (e.g., labeling a subtly negative statement as neutral), this trade-off is acceptable in applications prioritizing comprehensive aspect coverage,

such as business intelligence or user feedback analysis, where missing an aspect entirely is more detrimental than a slight sentiment error.

V. CONCLUSION

In this work, we studied the coverage–accuracy trade-off in aspect-level sentiment classification, especially for cases where the target aspect category is predefined but the corresponding sentiment evidence is expressed implicitly in the context.

To address this challenge, we proposed the **Aspect-guided Multi-head Cross-Attention (AMCA)** model, an end-to-end framework that uses learnable aspect embeddings as queries in a multi-head cross-attention mechanism over PLM-encoded contextual representations. This design allows the model to dynamically capture aspect-relevant contextual cues without relying on external tools or auxiliary sentences.

Extensive experiments on SemEval-2014 and SentiHood show that AMCA achieves competitive or improved F1 scores across several standard PLM backbones, including BERT, RoBERTa, and BART. The gains vary across architectures and metrics: for RoBERTa-based models, improvements are more recall-oriented, whereas for BERT- and BART-based models, improvements are mainly reflected in precision or overall F1 balance. Ablation studies confirm the contribution of cross-attention, multi-head diversity, and learnable aspect embeddings. Qualitative analysis further shows that AMCA can associate implicit contextual cues, such as “the wait was unacceptably long,” with the predefined service aspect and predict the corresponding sentiment.

Theoretical contribution. This work highlights the coverage–accuracy trade-off as an important evaluation perspective in ASC, suggesting that a slight decrease in some accuracy metrics may accompany improved modeling of difficult implicit aspect-related cases.

Practical contribution. AMCA offers a lightweight, end-to-end solution that can be integrated into existing PLM pipelines without external dependencies, making it suitable for applications where comprehensive aspect-level sentiment coverage is important.

Limitations. First, evaluation is limited to two datasets; extending to more domains would strengthen generalizability. Second, AMCA’s benefits diminish on highly specialized models that already incorporate aspect-specific pretraining.

Third, because the model assumes predefined aspect categories, it does not perform open-ended aspect term extraction.

Future work. We plan to integrate external knowledge for enhanced implicit reasoning, explore multi-modal cues, and extend AMCA to settings that jointly model aspect extraction and sentiment classification.

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